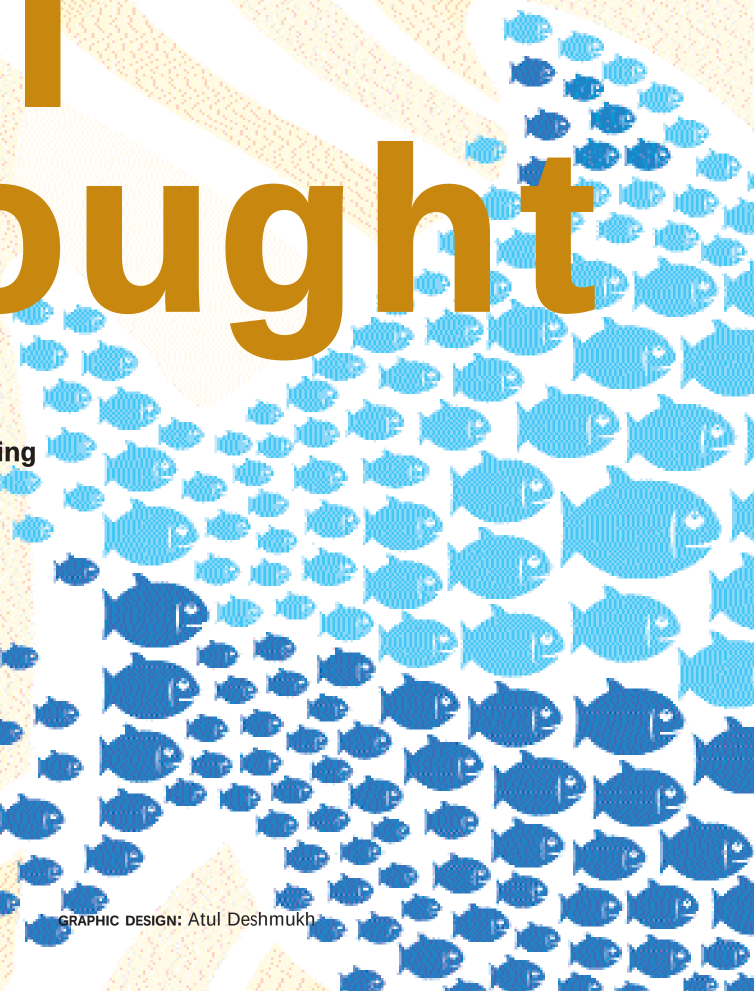
A school of thought

By the masses, for the masses: we take a look at a technology that promises to democratise computing

In India, as in most parts of the world, computers are a significant investment for ordinary people. One that involves bargain hunts for the cheapest hardware that money can buy and a muted exchange of illicit software that money need not. From the collective mind of technology pioneers around the world has emerged a concept that promises to change the landscape of computing. From something that only a handful of people, blessedly within a particular income bracket can make use of, computing hopes to achieve a utilitarian status—becoming as ubiquitous as electricity. Such is the promise of grid computing: a delivery system of computer resources much similar to an electric outlet. Just throw a switch and get instant, hassle-free access to computer power. Cheap too.

Come together

To better understand this concept, we need to take a closer look at the analogy of an electric power grid. Such a grid gathers current from resources that range from hydroelectric dams to nuclear reactors. Then using a vast network of cables, capacitors and transformers, it distributes the gathered power to homes and institutes geographically distant. Similarly, a computing grid will gather geographically distributed resources of CPU power, storage capacities, network bandwidth, printers, people and instruments and offer the collective under a singular, OS-independent banner to those who need it. There is strength in unity, computers subscribe to this law as well. Multiple heads performing a task are much better than one—connect two computers and the resultant power increases. Imagine connecting supercomputers in such a way, or connecting the millions of desktop computers that are equivalent in performance to a supercomputer of yesteryear. The computing power thus unleashed could be of staggering proportions. All you would need to do is plug into this hive of unified strength and draw transparently upon resources hitherto out of your reach.



GRAPHIC DESIGN: Atul Deshmukh

Now that we have a logical structure of a grid in mind, let us take a brief look at what makes such a network tick. A grid is a peer-to-peer network comprising four layers. The lowest rung or layer 1 is the heart of the unit. Called the Fabric, it comprises physical devices and resources that will be shared across the network—CPUs, storage devices, bandwidth, applications and peripherals, to name a few. Above lies layer 2, the Connectivity and Resource chunk comprising communication and authentication protocols that allow sharing of data and resources from layer 1. The Collective layer or layer 3 consists of services and APIs that allow transparent interaction between resources. Finally, user applications form the topmost layer 4. Just as a Web browser can access data from the WWW via a defined set of standards and protocols, so will applications sitting on this layer make use of underlying structures to access the resources of layer 1.

As an example, consider that one of your applications (sitting on layer 4) wishes access to a database (sitting on layer 1) over a grid. It would first establish its credentials for a secure connection via the